Exemplos

1.

$$\lim_{x\to 4}\frac{x^2-16}{x-4}=\lim_{x\to 4}\frac{(x-4)(x+4)}{x-4}$$

$=\lim_{x\to 4}(x+4)=8$;$x^2 - a^2=(x-a)(x+a)$

2.Resolvam

$$\lim_{x\to 5}\frac{x^2-25}{x-5}$$

$\lim_{x\to 5}\frac{(x-5)(x+5)}{x-5}=\lim_{x\to 5}(x+5)=10$

3.$\lim_{x\to 2}\frac{x^2-4}{x^2-3x+2}$ = 0/0 ; $ax^2+bx+c = a(x-x_1)(x-x_2)$

$=\lim_{x\to 2}\frac{(x-2)(x+2)}{(x-2)(x-1)}=$

$\lim_{x\to 2}\frac{x+2}{x-1}=\frac{4}{1}=4$

4.$\lim_{x\to 2}\frac{x^3-3x^2-x+6}{x-2}$

a) $(x^3 - 3x^2 - x + 6) : (x - 2)$

$$
\begin{array}{l|l}
x^3 - 3x^2 - x + 6 & x - 2 \\
\cline{2-2}
-x^3 + 2x^2 & x^2 - x - 3 \\
\cline{1-1}
0 - x^2 - x + 6 & \\
\quad x^2 - 2x & \\
\cline{1-1}
\quad 0 - 3x + 6 & \\
\qquad 3x - 6 & \\
\cline{1-1}
\qquad 0 &
\end{array}
$$

Assim

$X^3-3x^2-x+6 = (x-2)(x^2-x-3)$ e no limite

$$\lim_{x\to 2} \frac{x^3-3x^2-x+6}{x-2} = \lim_{x\to 2} \frac{(x-2)(x^2-x-3)}{x-2} =$$

$$\lim_{x\to 2}(x^2 - x - 3) =$$

$4 - 2 - 3 =$

-1

5. $\lim_{x \to 3} \frac{9x^3 - 36x^2 + 29x - 6}{x - 3}$

Dividindo

$$
\begin{array}{l|l}
9x^3 - 36x^2 + 29x - 6 & x - 3 \\
\hline
-9x^3 + 27x^2 & 9x^2 - 9x + 2 \\
\hline
\quad -9x^2 + 29x - 6 & \\
\quad\quad 9x^2 - 27x & \\
\hline
\quad\quad\quad 2x - 6 & \\
\quad\quad\quad -2x + 6 & \\
\hline
\quad\quad\quad\quad 0 &
\end{array}
$$

$$9x^3 - 36x^2 + 29x - 6 = (x - 3)(9x^2 - 9x + 2)$$

$$\lim_{x \to 3} \frac{9x^3 - 36x^2 + 29x - 6}{x - 3} =$$

$$\lim_{x \to 3} \frac{(x - 3)(9x^2 - 9x + 2)}{x - 3} =$$

$$\lim_{x \to 3} (9x^2 - 9x + 2) =$$

81-27+2=56

6. $\lim_{x\to 9} \frac{\sqrt{x}-3}{x-9}$ =? (a-b)(a+b) = a^2-b^2

$$\lim_{x\to 9} \frac{(\sqrt{x}-3)(\sqrt{x}+3)}{(x-9)(\sqrt{x}+3)}$$

$$(\sqrt{x}-3)(\sqrt{x}+3) = x + 3\sqrt{x} - 3\sqrt{x} - 9 = x - 9$$

$$\lim_{x\to 9} \frac{x-9}{(x-9).(\sqrt{x}+3)} = \lim_{x\to 9} \frac{1}{(\sqrt{x}+3)} = \frac{1}{6}$$

7. Calcule

$$\lim_{h\to 0} \frac{\sqrt{1+h}-1}{h}$$

$$\frac{(\sqrt{1+h}-1)}{h}\frac{(\sqrt{1+h}+1)}{(\sqrt{1+h}+1)} = \frac{(1+h-1)}{h(\sqrt{1+h}+1)} = \frac{1}{(\sqrt{1+h}+1)}$$

Assim

$$\lim_{h\to 0} \frac{\sqrt{1+h}-1}{h} = \lim_{h\to 0} \frac{1}{(\sqrt{1+h}+1)} = \frac{1}{2}$$